

Teaching notes on exploring the impact of AI on Supply Chain Management

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Abstract: Supply Chain Management (SCM) lays at the heart of every functioning business. The advent of digital technologies and especially big data and artificial intelligence (AI) have revolutionized the three main pillars of SCM: procurement, operations and logistics. This paper is a collection of theory, real world examples, methods and open research issues on the impact of AI on SCM, comprising a comprehensive set of teaching notes for those who choose to study and research this multi-disciplined topic that cross-cuts business and technology sciences.

Keywords: supply chain management, procurement, operations, logistics, artificial intelligence, big data, digital value creation

I. Introduction

“Operations and supply chain management is a key element in the improvement in productivity in business around the world.” (Jacobs et. al., 2011) Supply Chain Management (SCM) has been recognized as a key competence nowadays, heavily leveraged by digital technologies, which contributes significantly to corporate strategic capabilities and resilience. For example “IKEA’s entire integrated process, including the design of products, design of the packaging, manufacturing, distribution and retail outlets, is carefully wired toward delivering functionally innovative products at the lowest cost possible.” (ibid).

Companies that excelled in SCM, such as Amazon and ZARA / inditex group, figure as top global performers in their sectors. According to the Council of Supply Chain Management Professionals, SCM is defined as “The planning and management of all activities involved in sourcing and procurement, conversion, and all logistics management activities ... includes coordination with channel partners, which can be suppliers, intermediaries, third-party service providers, and customers.” (Wisner et. al., 2023) According to the Institute for Supply Management, SCM is “the design and management of seamless, value-added processes across organizational boundaries to meet the real needs of the end customer.” (ibid)

As a machine learning based technology, the effectiveness and trustworthiness of AI depends heavily on big data, which are characterized by volume, veracity, velocity, variety and value (Stair and Reynolds, 2021). Data produced by SCM daily work qualify as a source of big data, because SCM is broadly comprised of the main pillars: procurement, operations and logistics, all three of which generate data constantly, as long as a corporation is doing business.

As “42% of firms adopting AI tools reported cost savings¹”, it is obvious that AI has become a valuable tool to raise SCM efficiencies and effectiveness. This short teaching notes paper examines the impact of digital technologies and AI onto these three distinct pillars (procurement, operations and logistics) to demonstrate their power in digitally transforming the business processes, the methods of work and the means of performing SCM activities in the 21st century, which have become fundamentally different from those employed in the previous century, allowing orders of magnitude in improvements of the basic

¹ <https://www.scmr.com/article/cognitive-procurement-with-chatbots-pros-and-cons>

performance indicators of modern businesses. Some of the material presented below is based on interactions with our undergraduate students in relevant courses taught in the Business and Technology Department of Webster University, Athens campus.

II. The impact of AI on procurement

Procurement that involves sourcing, purchasing and managing the goods and services a company needs, is undergoing major changes due to AI. Traditionally, procurement requires a lot of manual work, such as supplier research, contract negotiations and invoice processing. However, as supply chains become larger, faster and more global, traditional methods are no longer enough. (Baryannis et al., 2019)

AI offers new ways to support procurement teams by handling large amounts of data, predicting risks, automating routine tasks and improving accuracy. As a result, procurement does not only have an operational function, but it is also becoming a strategic tool. While AI is transforming procurement by enhancing accuracy, intelligence, prediction and long-term planning, the European Commission (2018) has pointed out the importance of transparency and human control on AI systems, since SCM is an integral part of any corporate governance system. Predictive Analytics (PA) is a valuable source for demand planning and forecast data that drive procurement decisions. Although PA is not exactly AI, it is the appropriate precedent for enabling organizations to make more accurate proactive decisions (Balkan and Akyuz, 2025) that will feed AI with input to initiate the procurement business process according to the procurement strategies defined.

AI can analyze data, calculate optimal pricing, collect corporate supplier sanity news and evaluate supplier performance metrics, classify spending, detect unusual purchasing patterns, and highlight potential savings more accurately and quickly than manual methods. Researchers also note that AI can help streamline sourcing by interpreting user needs, comparing supplier proposals, and assisting with basic contract drafting. In procurement, AI enhances efficiency, improves decision-making, and enables procurement teams to allocate their time and resources towards more value-added activities. Areas of procurement that remain human benefit include intuition, negotiations, supervision and relationship management. The area of devising new procurement strategies is still a human dominion, but AI is quickly catching up in creativity, using new inference capabilities, especially when enough training material is shaped to get insights about the different procurement strategies, such as those explained by the Kraljic matrix (see for example Caniels & Gelderman, 2005).

If both buyers and suppliers were to use AI for automatic and autonomous negotiations², a series of issues – probably risks – could emerge: lack of consensus, power imbalances (if one AI is more advanced than another), potential data manipulation, potential damage on bilateral relationships, uncertainty about regulatory oversight, to name a few. To start exploring these areas, the use of digital twins in procurement³ is a good idea since it offers a safe environment for inventions. That is a virtual copy of procurement processes, suppliers, markets and negotiations. A procurement digital twin consists of virtual supplier profiles, AI prediction machines, historical and performance data, market data feeds, contract and negotiation models. Unlike traditional AI tools that automate tasks, a procurement digital twin allows companies to test decisions before acting in the real world. In procurement, this means companies can virtually simulate contract negotiations, predict supplier risks and examine the effects of

² <https://www.scmr.com/article/the-invisible-handshake-ai-to-ai-procurement-negotiations>

³ <https://www.scmr.com/article/how-digital-twin-technology-can-revolutionize-a-supply-chain>

market changes. Interested readers may find on-going research in this area in Ike et. al., (2025), Meschini, et. al., (2022), Ravikumar & Aarthi (2026) etc.

For all these reasons, it is right to claim that AI is redefining the procurement function. (Allal-Chérif et. al, 2021)

III. The impact of AI on operations

A McKinsey report (2022) found that 56% of companies surveyed had already implemented AI in at least one area of their operations, and another 25% planned to do so within the next two years. This trend highlights the critical role AI plays in shaping operational strategies and maintaining a competitive edge.

The evolution of AI in operations has been accelerated by advances in machine learning, predictive analytics, robotic process automation (RPA) (Lacity & Willcocks, 2016), and natural language processing (NLP). Organizations now have the ability to not only automate routine tasks but also to gain insights from massive datasets, enabling more informed, and proactive decisions. For instance, AI systems can predict machinery breakdowns before they occur, optimize inventory levels, and even identify inefficiencies in supply chains.

AI in Manufacturing Operations: Operations management, which involves planning, organizing, and supervising production, workflow, and service delivery, is increasingly dependent on AI technologies to maintain competitive advantage in a fast-paced market (Choi et. al., 2018). It's been reported that predictive maintenance can reduce unplanned downtime by up to 30% and maintenance costs by 20%, significantly improving overall productivity. In addition, AI-driven quality control systems, using computer vision and machine learning, can detect defects in real time. For example, automotive manufacturers use AI-powered vision systems to inspect components on assembly lines, reduce human error and ensure consistent quality standards (Ivanov et. al., 2019).

AI in Service Operations: Beyond manufacturing, AI has made significant inroads into service operations. In healthcare, AI applications in scheduling, patient management, and diagnostic support have streamlined operations and improved service delivery. Hospitals using AI for resource allocation have reported shorter patient wait times and more efficient use of staff (Makridakis, 2017). In retail and e-commerce, AI helps optimize inventory management by predicting demand fluctuations and recommending stock adjustments. For example, Amazon uses AI algorithms to forecast product demand, ensuring products are available while minimizing overstocking.

Challenges in AI Adoption: despite the benefits, organizations face challenges in AI implementation. High costs, lack of skilled personnel, data privacy concerns, and resistance to change are commonly cited barriers (Brynjolfsson & McAfee, 2017). Additionally, integrating AI with legacy systems can be technically complex. Nevertheless, firms that invest strategically in AI often experience significant returns in efficiency and operational resilience.

Steps in adopting AI in SCM operations

(a) **Target tasks and processes:** Organizations must first identify operational processes that are repetitive, data-intensive, or prone to errors. Examples include scheduling, inventory management, procurement tasks, and quality inspections. These processes are prime candidates for AI automation. RPA in administrative workflows can reduce task completion times by 50% (Lacity & Willcocks, 2016).

(b) Data Collection and Preparation: High-quality data is essential for AI effectiveness. Historical and real-time operational data should be collected, cleansed, and structured to ensure accurate AI predictions. Data may include machine sensor readings, production logs, inventory records, and customer service metrics.

(c) AI Technology Selection: Selecting the appropriate AI technology depends on the operational need:

- Machine Learning: For predictive maintenance, demand forecasting, and anomaly detection.
- Robotic Process Automation (RPA): For automating repetitive administrative tasks, such as invoicing or report generation.
- Natural Language Processing (NLP): For customer service automation, chatbots, and document processing.

(d) Model Development and Testing: AI models are trained using collected data and evaluated using metrics such as accuracy, precision, and recall. Pilot testing in a controlled environment ensures the AI system performs reliably before full deployment.

(e) Deployment and Monitoring: After deployment, continuous monitoring and periodic retraining of AI models are essential. Performance metrics such as process efficiency, error rates, cost savings, and throughput should be tracked to measure success. For example, energy-intensive operations in manufacturing can be adjusted based on AI recommendations, reducing energy costs by up to 20% (Makridakis, 2017).

IV. The impact of AI on logistics

To many people, logistics means transportation; which is partly true, because it is much more than this. However, transportation is a good starting point for AI adoption. AI models for optimization strategies on transportation (Afolayan et. al., 2024) seem to be an effective evolution from the traditional operational research methods. Major companies such as UPS, DHL, and Amazon have publicly stated that AI-based route optimization has saved them millions of dollars each year. For example, UPS built ORION⁴ to “optimize delivery routes across millions of packages and thousands of drivers. By using real-time data, AI, and logistics math to find the most efficient route daily, ORION saves 100+ million miles, \$300M+ in cost, and thousands of tons of CO₂ annually.”

Another significant area discussed in the literature is warehouse automation. Sodiya et. al. (2024) describe how “AI-powered robotics and automation technologies enhance the efficiency warehouse automation”, while other writers emphasize on AI-powered robots that are commonly found nowadays in most of the major distribution centers. These robots can move items, sort packages, and assist with picking tasks. Moreover, they have visit specific stocked packages to collect RFID-based information regarding the conditions of the saved materials. Their accuracy reduces human errors and improves overall efficiency. Other authors, for example Mehmi et. al. (2018). Le-Anh & De Koster (2006) highlight the rising use of Automated Guided Vehicles (AGVs), which transport materials inside warehouses without human involvement. These systems rely on AI algorithms to navigate safely around obstacles, making them reliable even in fast-paced warehouse environments. IBM⁵ argues that AI-enabled supply chain visibility tools have become essential for detecting delays, shortages, and other disruptions in real time. These

⁴ dev.webopstechnologies.com/wp-content/uploads/2025/05/UPS.pdf

⁵ [The intuitive, AI-powered supply chain | IBM](#)

studies show that AI is not only improving operational performance but also strengthening long-term supply chain resilience.

A systematic literature review AI and big data analytics for supply chain resilience is given by Zamani et al. (2023)

V. Some conclusions

The use of AI in SCM has already resulted in improvements in operational efficiency, cost reduction, enhanced decision making, better customer experience (e.g. using chat-bots for pre-sale and after-sales support), risk management etc. Investments in all three main pillars of SCM, procurement, operations and logistics are costly but necessary to reap the anticipated benefits. As global supply chains continue to become more complex, AI will play an even more significant role in ensuring that logistics systems remain efficient, scalable, and resilient. With digital technologies continually evolving, AI is expected to play an even greater role in logistics through advanced robotics, autonomous delivery systems, and predictive intelligence, at both sides, supplier and buyer, since one party usually drives the other to more innovative approaches. Future research may examine how AI integrates with emerging technologies such as block-chain and the Internet of Things to build even smarter logistics networks.

Effective adoption of AI should be based on quality information, integration with enterprise information systems, training of employees, and ethical leadership. Human supervision and ultimate decision making remains to be of paramount importance, since, after all, AI is a tool, not the businessman. Measuring the value of AI beyond cost reductions is a more demanding research task. A rigorous methodological approach to the analysis of e-procurement benefits is proposed by Trkman & McCormack (2009), but more research approaches are needed in order to justify the benefits for such huge investments.

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